Subsidizing Climate Risk: Flood Insurance, Property Values, and Moral Hazard in England

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Abstract

Government-subsidized flood insurance shields homeowners from climate risk, but whether subsidies capitalize into property values remains poorly understood. I exploit a triple-difference design combining the UK's Flood Re reinsurance scheme—which caps premiums for properties built before 2009 but excludes newer construction—with postcode-level flood risk classifications and 13 million English property transactions. The most saturated specifications yield DDD estimates of −1.4% to −1.8%, providing no evidence that Flood Re eligibility boosted property values. However, placebo tests in low-risk areas—where the subsidy is irrelevant—produce effects of similar magnitude, indicating that the estimates partly reflect vintage depreciation trends in flood-classified areas rather than the insurance channel alone. These results provide evidence against large positive capitalization but cannot isolate the Flood Re mechanism from confounding vintage trends.

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1. Introduction

Two houses on a flooded street in Northern England may look identical—same terrace design, same postcode, same risk of the river entering the living room. But if one was built in December 2008 and the other in January 2009, their insurance bills differ by thousands of pounds. The older house qualifies for the UK's Flood Re scheme, which caps its flood insurance at a few hundred pounds annually; the newer house does not, leaving its owner to face a private market where premiums can exceed £5,000. This paper asks whether that annual "insurance tax" shows up in home prices.

This paper asks whether and how subsidized flood insurance affects housing markets. The question matters for three reasons. First, flood insurance subsidies are enormous—the US National Flood Insurance Program has accumulated over $20 billion in debt to the Treasury, and the UK's Flood Re scheme processes over £200 million in premiums annually (Kousky and Michel-Kerjan, 2017; Flood Re Ltd, 2023). Second, the capitalization of insurance subsidies into property values has direct implications for the incidence of these programs: if subsidies are passed through to prices, they benefit sellers rather than improving affordability for buyers. Third, the moral hazard implications are urgent: if subsidized insurance makes flood-zone living artificially cheap, it works directly against climate adaptation goals like managed retreat from at-risk areas (DePietri and McPhearson, 2022).

Identifying the causal effect of insurance subsidies on property values is difficult because flood risk, insurance availability, and property values are simultaneously determined. Homes in flood zones may be cheap because of flood risk, or because of correlated neighborhood characteristics, or because of insurance costs—and separating these channels requires an exogenous shock to insurance costs that does not directly affect flood risk or other property attributes.

I exploit a unique institutional feature of the UK's Flood Re reinsurance scheme to overcome this identification challenge. Flood Re, launched in April 2016, caps the cost of flood insurance premiums for residential properties based on Council Tax band, making flood insurance "available and affordable" in high-risk areas. Crucially, Flood Re explicitly excludes properties built on or after 1 January 2009. This exclusion was designed to discourage new construction in flood zones—but it also creates a sharp within-flood-zone comparison group. Properties built before 2009 in a flood zone receive the insurance subsidy; properties built after 2009 in the same flood zone, facing identical flood risk, do not. This motivates a triple-difference (DDD) design that compares the price evolution of eligible versus ineligible properties, in flood zones versus non-flood zones, before versus after Flood Re's launch.

The DDD design offers three advantages over previous approaches. First, it nets out flood-

zone-specific time trends unrelated to insurance—such as changing climate risk perceptions, infrastructure investment, or regulatory changes—that could confound a simple before-after comparison within flood zones. Second, it nets out any construction-vintage-specific price trends that might differ between older and newer properties nationwide. Third, it provides a built-in placebo: properties in very-low-risk areas, where Flood Re is irrelevant because standard insurance is already affordable, should show no DDD effect.

I combine two large administrative datasets covering the universe of relevant transactions: the HM Land Registry Price Paid Data, which records every residential property transaction in England and Wales since 1995 (over 24 million transactions), and the Environment Agency's Open Flood Risk by Postcode dataset, which classifies every English postcode into flood risk bands (High, Medium, Low, Very Low). The Land Registry data includes a "New Build" flag that identifies construction vintage, enabling the eligibility classification central to the DDD.

The main finding is that the data provide no evidence of positive capitalization. The most credible specifications—with district-by-year-quarter or postcode-level fixed effects—yield DDD estimates of -0.014 (SE: 0.003) and -0.018 (SE: 0.004), respectively. A naive specification without property controls produces a positive estimate of $+0.094$, but this reverses once compositional differences between pre-2009 and post-2009 properties are absorbed. Crucially, placebo tests using Very Low risk areas—where Flood Re is irrelevant—reveal a significant negative DDD of -0.134, comparable in magnitude to the main effect. This indicates that the DDD partly captures vintage-related trends in flood-classified areas rather than the insurance subsidy specifically. The event study shows pre-existing trends (joint $p = 0.011$), and HonestDiD sensitivity analysis confirms that confidence intervals include zero under moderate violations of parallel trends. These diagnostics mean the results should be interpreted as evidence against large positive capitalization rather than as a clean causal null.

I investigate two auxiliary outcomes. First, transaction volumes in flood zones slightly declined after 2016 (-0.6%, SE: 0.1%), contrary to the hypothesis that Flood Re would restore market liquidity. Second, the share of new construction in flood zones fell by 4.9 percentage points (SE: 0.8%), suggesting that the market did partially internalize Flood Re's exclusion of post-2009 properties—though the timing aligns with broader tightening of planning regulations in flood-risk areas.

This paper contributes to three literatures. First, I advance the literature on flood risk and property values (Bin and Polasky, 2004; Bernstein et al., 2019; Murfin and Spiegel, 2020; Barrage and Bakkensen, 2017; Mayson and Sherrington, 2024) by providing the first triple-difference estimate that exploits within-flood-zone variation in insurance eligibility. While Garbarino et al. (2024) found a positive price effect using a simpler difference-in-differences design, the DDD—which nets out concurrent flood-zone trends—finds no positive

capitalization and reveals that the DiD estimate may conflate insurance effects with correlated shocks. The DDD itself is contaminated by vintage depreciation trends, but it establishes an upper bound on positive capitalization that is informative for policy. Second, I contribute to the literature on moral hazard in disaster insurance (Hudson et al., 2017; Browne and Hoyt, 2000; Kunreuther, 2015) by showing that new construction in flood zones declined after the Flood Re exclusion, suggesting at least partial market internalization of eligibility rules. Third, the results are policy-relevant for the climate adaptation literature (DePietri and McPhearson, 2022; Deryugina, 2017): the evidence against large positive capitalization suggests that the political economy barriers to risk-reflective pricing transitions may be weaker than feared, though this conclusion is tempered by the identification challenges documented here.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the institutional background of Flood Re. Section 4 presents the conceptual framework. Section 5 describes the data. Section 6 details the empirical strategy. Section 7 presents results. Section 8 discusses implications, and Section 9 concludes.

2. Related Literature

This paper sits at the intersection of three literatures: the capitalization of environmental risk into property values, the design and effects of natural disaster insurance, and the political economy of climate adaptation.

2.1 Flood Risk and Property Values

A large hedonic literature documents that flood risk is negatively capitalized into property prices, though estimates vary considerably with context and salience. Bin and Polasky (2004) estimate a 5–10% price discount for properties in FEMA-designated floodplains in North Carolina, while Bernstein et al. (2019) show that properties exposed to sea-level rise trade at a 7% discount relative to otherwise comparable properties, with the discount growing over time as climate awareness increases. Murfin and Spiegel (2020) find that the sea-level-rise discount is absent in markets with less climate-aware populations, highlighting the role of information and attention. Koster and van Ommeren (2021) and Ortega and Taşpınar (2018) study flood risk capitalization in the context of major hurricane events, finding substantial but heterogeneous price effects that depend on salience, spatial proximity, and housing market conditions.

In the UK context, Mayson and Sherrington (2024) provides evidence that flood risk is capitalized into property values, with substantial variation across regions and risk levels.

Lamond and Proverbs (2009) document that flood-affected properties in England suffer persistent price discounts of 12–20% even years after flood events, suggesting that flood risk information is sticky in local markets. More recently, Garbarino et al. (2024) directly study the Flood Re scheme using a difference-in-differences design comparing flood-zone and non-flood-zone properties before and after the scheme's launch. They find a positive price effect of approximately 1.5%, which they attribute to improved insurance availability reducing the effective cost of flood-zone homeownership. My triple-difference design extends their work by netting out this flood-zone-specific trend using within-flood-zone variation in Flood Re eligibility.

The literature on disaster salience and risk updating is also relevant. Gallagher (2014) shows that flood insurance take-up spikes after local flood events but decays rapidly, suggesting that risk perceptions are driven more by recent experience than objective probability. Atreya et al. (2013) find that flood risk discounts in the US housing market erode substantially within 4–9 years of a flood event, consistent with "forgetting" or recency bias. Gibson and Mullins (2019) document similar attention dynamics in real estate markets. These findings imply that the timing of Flood Re's launch—months after severe December 2015 floods—may have interacted with heightened flood risk salience in ways that confound the DDD if salience effects differ between pre-2009 and post-2009 properties.

2.2 Natural Disaster Insurance Design

The economics of disaster insurance centers on the tension between risk-sharing and moral hazard. Kunreuther (2015) provide a comprehensive framework showing that subsidized insurance encourages development in hazardous areas, underprices risk, and creates fiscal liabilities. The US National Flood Insurance Program (NFIP) is the canonical example: Kousky and Michel-Kerjan (2017) document that NFIP subsidies—which reduced premiums by 40–60% for "pre-FIRM" properties built before initial flood maps—contributed to over $20 billion in Treasury debt and concentrated losses in repetitive-loss properties.

Several papers study whether NFIP reforms have affected property values. Bin and Landry (2008) find that the Biggert-Waters Act of 2012, which mandated a transition to actuarial pricing, reduced property values in flood zones by 2–5% before political pressure led to its partial repeal. Browne and Hoyt (2000) examine the broader relationship between catastrophe insurance pricing and property development, finding that subsidized rates encourage construction in at-risk areas. Flood Re differs from the NFIP in a crucial design feature: the pre-2009 eligibility cutoff explicitly prevents subsidies from encouraging new construction, addressing the moral hazard concern at the program's inception rather than retroactively.

2.3 Climate Adaptation and Managed Retreat

The broader policy motivation for understanding insurance capitalization comes from the climate adaptation literature. DePietri and McPhearson (2022) argue that subsidized insurance, by making flood-zone living artificially affordable, works against managed retreat—the planned relocation of communities away from high-risk areas. Deryugina (2017) show that federal disaster transfers generate large fiscal externalities across jurisdictions, as governments transfer risk from local communities to national taxpayers. If subsidized insurance capitalizes into property values, it creates a concentrated constituency of homeowners who would suffer capital losses from any reform, generating political economy barriers to risk-reflective pricing transitions (Hsiang et al., 2017).

The results in this paper speak to this debate, albeit with caveats about identification. The absence of evidence for large positive capitalization suggests that the transition to risk-reflective pricing—whether through Flood Re's planned phase-out in 2039 or through NFIP reform—may face less political opposition than feared, since current homeowners' wealth does not appear to be substantially inflated by the subsidy. However, the confounding vintage trends documented here mean this conclusion should be treated as suggestive rather than definitive.

3. Institutional Background

3.1 Flood Risk in England and Wales

England has approximately 5.2 million properties at some level of flood risk, with roughly 1.4 million at significant risk (greater than 1-in-75 annual probability). The Environment Agency classifies flood risk from rivers and the sea into four bands: High (>3.3% annual probability), Medium (1–3.3%), Low (0.1–1%), and Very Low (<0.1%). These classifications account for existing flood defenses and their condition, reflecting residual risk after mitigation infrastructure. The classifications are publicly available at the postcode level, allowing both homebuyers and lenders to assess flood exposure before transactions.

The geographic distribution of flood risk in England is highly concentrated. The highest-risk postcodes are predominantly located along major river floodplains (Thames, Severn, Trent, Ouse), in coastal areas subject to tidal flooding (East Anglia, Humber estuary, Somerset Levels), and in low-lying urban areas with surface water drainage vulnerabilities (parts of London, Leeds, Sheffield). This geographic concentration means that a relatively small number of local housing markets bear the brunt of flood risk, and policies affecting flood insurance have outsized effects in these markets.

Flooding imposes substantial and recurring costs. Major flood events—notably in 2007 (Yorkshire, Midlands), 2013–14 (Somerset, Thames), and 2015–16 (Cumbria, Lancashire, Yorkshire)—caused billions of pounds in damage and displaced thousands of households. The December 2015 floods, which occurred just months before Flood Re's launch, caused over £1.6 billion in insured losses and reinforced the political urgency of addressing flood insurance affordability. Importantly, flood damage is not limited to high-risk areas: surface water flooding from intense rainfall can affect properties in any flood risk band, though the probability and severity of flooding increases monotonically with the risk classification.

The expected cost of flood damage is projected to rise substantially under climate change scenarios. The Committee on Climate Change estimates that without additional adaptation measures, annual expected flood damage in the UK could increase by 50–100% by 2050 under a 2°C warming pathway, and by up to 200% under higher emissions scenarios. This trajectory makes the design of flood insurance institutions—and their interaction with property markets and development incentives—a question of growing urgency.

3.2 The Pre-Flood Re Insurance Market

Before Flood Re, flood insurance in the UK operated under a voluntary "Statement of Principles" (SoP) between the government and the Association of British Insurers (ABI). First agreed in 2000 and renewed in 2008, the SoP committed insurers to continue offering flood cover to existing domestic customers in areas where the government was investing in flood defenses. In return, the government committed to increasing expenditure on flood defense infrastructure and improving planning controls to reduce development in flood-risk areas (Department for Environment, Food and Rural Affairs, 2012).

But this arrangement was breaking down. As climate change increased flood frequency and severity, insurers raised premiums sharply or withdrew cover entirely from high-risk areas. The ABI estimated that by 2013, approximately 200,000 households could not obtain affordable flood insurance, with quoted premiums for some high-risk properties exceeding £5,000–£10,000 annually. In extreme cases, insurers refused to offer cover at any price. The SoP's protections applied only to existing customers, not to new purchases, creating a particularly severe problem for market transactions: a homeowner with affordable cover through a legacy insurer would lose that protection upon selling, because the buyer would face the open market.

The resulting "insurance gap" created severe hardship for flood-zone homeowners and had important consequences for housing markets. Properties that could not obtain affordable insurance became effectively unmortgageable: since UK mortgage lenders universally require buildings insurance that includes flood cover, the absence of affordable insurance functioned

as a binding constraint on transactions. Prospective buyers could not secure mortgages, trapping existing owners in depreciating and illiquid assets. Estate agents in flood-prone areas reported that properties could sit on the market for years, and that achieved prices were 20–30% below similar non-flood-zone properties—a discount well in excess of what flood risk alone would justify. The market failure was particularly acute in areas of high flood risk where the Statement of Principles provided no guarantee of continued cover, and where the December 2013–14 and 2015–16 flood events had been most damaging.

The political economy response to the insurance gap was swift. The Flood Insurance Working Group, established jointly by Defra and the ABI in 2011, considered several alternatives including a fully risk-reflective market (rejected as inequitable), a Pool Re-style government backstop (considered too costly), and what eventually became the Flood Re model—a cross-subsidy scheme funded by a levy on all home insurance policies. The Water Act 2014 provided the legislative framework, and Flood Re began operating on 4 April 2016.

3.3 Flood Re: Design and Eligibility

Flood Re was established by the Water Act 2014 and began operating on 4 April 2016. It is a reinsurance scheme jointly created by the UK government and the insurance industry, designed to make flood insurance "available and affordable" for residential properties in high-risk areas.

The scheme works by allowing participating insurers to "cede" the flood risk portion of household policies to Flood Re at a fixed premium based on the property's Council Tax band. As of 2023, these capped premiums range from £46 (Band A) to £924 (Band H). Flood Re then reinsures this pooled risk, funded by a levy on all UK home insurance policies (approximately £180 million annually). In effect, all UK homeowners cross-subsidize flood insurance for high-risk properties through a small addition to their premiums.

The critical eligibility rule: Flood Re covers only residential properties that were built and used as a dwelling *before* 1 January 2009. Properties built on or after that date are explicitly excluded, regardless of their flood risk. This cutoff was chosen to avoid creating perverse incentives for new construction in flood-prone areas. The rationale, stated in the Water Act's explanatory notes, is that new developments should "reflect the true cost of flood risk in their pricing" and that developers should be held responsible for building in appropriate locations.

Flood Re is designed as a transitional scheme. It was originally set to operate until 2039, with the expectation that the insurance market would gradually transition to risk-reflective pricing. Recent amendments may extend the scheme, but the pre-2009 eligibility cutoff remains fixed.

Aggregate evidence confirms that Flood Re substantially reduced premiums for eligible properties. By 2023, over 350,000 policies had been ceded to the scheme, and Flood Re reports that 97% of quotes for flood-risk properties became available at or below £1,250 annually, compared with the pre-scheme situation where many high-risk properties faced premiums of £5,000 or more. The ABI estimates average savings of £1,000–£2,000 for properties in the highest risk bands. While property-level premium data are not available for this analysis, the aggregate evidence provides a credible first stage: Flood Re did meaningfully reduce insurance costs for eligible properties, meaning the null result on property values reflects non-capitalization of a real subsidy rather than absence of any subsidy to capitalize.

3.4 Implications for Identification

The pre-2009 eligibility cutoff creates a natural experiment within flood zones. In principle, properties built before 2009 in a flood zone benefit from capped insurance premiums, while properties built on or after 2009 in the same zone, facing identical flood risk, do not. Any differential price change between these groups after April 2016, relative to the corresponding differential in non-flood areas, can be attributed to the insurance subsidy.

In practice, the Land Registry data does not record true construction year. The available proxy is the New Build flag, which identifies a property's *first sale* as new construction. This means the "ineligible" group consists exclusively of first-sale new builds transacted on or after 1 January 2009, while the "eligible" group contains all other transactions—including resales of post-2009 properties whose New Build flag reverts to "N" at second sale.

The magnitude of this misclassification can be bounded. By 2025, approximately 1.8 million properties have been first sold as new builds since 2009. With average holding periods of 5–7 years, a growing share of the post-2009 stock enters the resale market each year: I estimate that by 2020, approximately 15–25% of all transactions classified as "eligible" in flood zones may actually involve post-2009 properties. This contamination grows over time, making the "eligible" and "ineligible" groups increasingly similar and mechanically compressing the DDD coefficient toward zero.

The direction of this bias deserves careful discussion. Under classical measurement error (random misclassification uncorrelated with other regressors), the bias is toward zero—conservative for testing positive capitalization. However, the misclassification is not purely random: it is systematically correlated with the new-build market segment, which has distinct cyclical properties, compositional characteristics, and price dynamics. New-build first sales may be subject to developer pricing strategies, Help-to-Buy effects, and compositional shifts (toward flats, toward smaller units) that differ systematically across flood and non-flood locations. This non-classical measurement error could generate spurious DDD effects of either

sign, meaning the "conservative bias" interpretation should be treated with caution. The DDD coefficient is better understood as estimating the price differential between new-build first sales and all other transactions within flood zones, which includes but is not limited to the insurance eligibility channel.

4. Conceptual Framework

4.1 Insurance Subsidies and Property Values

A standard asset pricing model predicts that the value of a property equals the discounted stream of net housing services minus expected costs, including insurance premiums. When Flood Re reduces insurance costs for eligible properties by ΔI per year, the present value of this subsidy is:

$$\Delta V = \sum_{t=0}^{T} \frac{\Delta I_t}{(1+r)^t} \approx \frac{\Delta I}{r} \tag{1}$$

where r is the discount rate and T is the policy horizon (until 2039). If annual insurance savings are £500–£2,000 for high-risk properties and the discount rate is 5%, the predicted capitalization is £10,000–£40,000. However, capitalization may be attenuated by uncertainty about Flood Re's continuation, incomplete awareness of the subsidy, or bargaining frictions in property transactions.

4.2 Partial Capitalization and Attenuation

Several factors may attenuate capitalization below the full present-value benchmark. First, *finite policy horizon*: Flood Re is legislated to operate until 2039, making the subsidy stream finite rather than perpetual. With a 5% discount rate and 23-year remaining horizon (as of 2016), the present value is $\Delta I \times \frac{1-(1+r)^{-23}}{r} \approx 13.5 \times \Delta I$, rather than $20 \times \Delta I$ under a perpetuity. Second, *policy uncertainty*: even within the legislated horizon, political or fiscal shocks could modify the scheme. If buyers assign probability π to early termination, expected capitalization falls to $(1-\pi) \times \Delta V$. Third, *transaction costs and search frictions*: Harding et al. (2003) show that capitalization of local amenities is incomplete in markets with significant search costs, as buyers cannot perfectly sort on all property attributes. Flood Re eligibility may be a low-salience attribute relative to location, size, and condition, leading to incomplete price adjustment.

Fourth, and perhaps most importantly, *information asymmetry* may limit capitalization. Flood Re eligibility is not recorded in Land Registry data or standard property listings. Buyers must independently determine whether a property was built before 2009—information

that is often ambiguous for properties that have been renovated or extended. Estate agents are not required to disclose Flood Re eligibility status. If a substantial fraction of buyers are unaware of or confused about the eligibility distinction, the subsidy cannot fully capitalize.

4.3 Moral Hazard and Development Incentives

Flood Re's designers anticipated moral hazard and built the pre-2009 cutoff to prevent it. The prediction is clear: if the market correctly prices the ineligibility of new properties, developers should face lower demand for flood-zone construction and reduce building in at-risk areas.

The moral hazard channel operates through the development decision. Consider a developer choosing between a flood-zone site and a non-flood-zone site. Before Flood Re, the insurance cost for buyers in both locations reflects market pricing. After Flood Re, the insurance cost for existing (pre-2009) properties in the flood zone is capped, but the cost for new builds remains market-priced—and may even increase as insurers adjust to the scheme's risk reallocation. The developer internalizes that buyers of new flood-zone properties face a competitive disadvantage relative to buyers of existing stock, reducing expected sale prices and development profitability.

However, moral hazard may persist through several channels. First, buyers may not know about or understand the eligibility rule, blunting its incentive effects. Second, developers may gamble that political pressure will extend the cutoff date, particularly if housing supply shortages intensify. Third, the existence of Flood Re may signal general government willingness to intervene in flood-risk markets, reducing perceived tail risk and encouraging development even in the absence of explicit insurance coverage.

4.4 Testable Predictions

The framework generates four testable predictions:

Prediction 1 (Capitalization): Eligible (pre-2009) properties in flood zones should appreciate relative to ineligible (post-2009) properties in the same zones after April 2016.

Prediction 2 (Dose-Response): The effect should be larger in higher-risk flood zones, where the insurance cost reduction is greater.

Prediction 3 (Regressivity): The effect should be larger for higher-value properties, since Flood Re premiums increase with Council Tax band but are capped below market rates.

Prediction 4 (Moral Hazard): If the market correctly prices ineligibility, new construction should decline in flood zones relative to non-flood zones after 2016.

5. Data

5.1 Land Registry Price Paid Data

To track how the market priced the 2009 eligibility cutoff, I use HM Land Registry's Price Paid Data (PPD), which records every residential property transaction in England and Wales at the sale price. The dataset covers January 1995 through the most recent available date (early 2025), comprising over 24 million transactions. Each record includes the transaction price, date, postcode, property type (Detached, Semi-detached, Terraced, Flat), tenure (Freehold or Leasehold), and crucially, a "New Build" flag indicating whether the property was newly constructed at the time of sale.

I restrict the sample to standard residential transactions (PPD Category A) with valid postcodes, positive prices, and known property types. The New Build flag is central to the identification strategy: properties first sold as new builds on or after 1 January 2009 are classified as Flood Re-ineligible; all other transactions (established properties or pre-2009 new builds) are classified as eligible.

A limitation of this classification is that the New Build flag only identifies the *first* sale of a newly constructed property. Subsequent resales of post-2009 properties are recorded with Old/New = "N" and would therefore be classified as "eligible" (pre-2009) in my framework. This means the "ineligible" group comprises exclusively first sales of new construction, while the "eligible" group includes both genuinely pre-2009 stock and resales of post-2009 properties. This measurement error attenuates the DDD toward zero if resales of ineligible properties are incorrectly assigned to the eligible group. The direction of bias is conservative: it makes finding a positive insurance capitalization effect *harder*, so the null result cannot be attributed to this classification issue.

5.2 Environment Agency Flood Risk Data

I obtain postcode-level flood risk classifications from the "Open Flood Risk by Postcode" dataset, which combines the Environment Agency's Risk of Flooding from Rivers and Sea (RoFRS) assessment with Royal Mail postcode data. Each postcode receives one of four flood risk classifications: High (>3.3% annual probability), Medium (1–3.3%), Low (0.1–1%), or Very Low (<0.1%).

For the main analysis, I define "flood zone" as postcodes classified as High or Medium risk. This corresponds to the properties where Flood Re is most likely to be binding—standard insurance may still be affordable at Low or Very Low risk levels. I test sensitivity to alternative definitions (High only; High + Medium + Low).

5.3 Linking Datasets

The two datasets are linked at the postcode level. Each Land Registry transaction is matched to its postcode's flood risk classification. Since the Environment Agency flood risk data covers only England (not Wales), I restrict the analysis sample to English postcodes. Welsh postcodes are identified using the first letter of the postcode (SA, SY, LL, LD, CF, NP, HR prefixes mapping to Welsh areas) and excluded from the sample. Flood Re applies to the whole of the UK, but without comparable Welsh flood risk classifications from Natural Resources Wales, including Welsh transactions would introduce systematic misclassification in the treatment variable. The merge rate for English postcodes exceeds 95%.

5.4 Sample Construction

The main analysis sample covers 2009–2025, beginning when post-2009 new builds first appear in the data. The 2025 data includes transactions through approximately March 2025 (partial year). I also use the full 2005–2025 window for extended pre-period robustness checks. The sample restriction to 2009 onward ensures that the pre-2009/post-2009 construction vintage comparison is meaningful: before 2009, all properties are eligible by construction.

The final analysis panel contains 12,951,289 English transactions spanning 2009–2025 in the main sample (approximately 17 million in the extended 2005–2025 sample). Of these, 287,799 transactions (2.2%) occur in postcodes classified as High or Medium flood risk—the "flood zone" used for the main analysis. This share is small but reflects the geographic concentration of significant flood risk. Approximately 89.2% of all transactions involve pre-2009 (Flood Re-eligible) properties and 10.8% involve post-2009 (ineligible) new builds.

The pre-2009/post-2009 split is relatively stable over the sample period, though the share of post-2009 new builds naturally increases over time as more housing stock is added. In 2009, post-2009 new builds constitute approximately 7% of transactions; by 2025, this share rises to roughly 12%. This gradual compositional shift is absorbed by the year-quarter fixed effects in the DDD specification.

5.5 Summary Statistics

Table 1 reports summary statistics by flood zone status and construction vintage for the full English sample of 12,951,289 transactions (regression samples are marginally smaller due to singleton fixed effects dropped by `fixest`). Several patterns are noteworthy. First, flood-zone properties are generally less expensive than non-flood properties, consistent with a baseline flood risk discount. Second, pre-2009 (eligible) properties differ from post-2009 (ineligible) properties in composition: newer builds are more likely to be flats and leasehold, reflecting

Table 1: Summary Statistics by Flood Risk and Construction Vintage

		Price (GBP)			Property Type (%)				Tenure (%)	
Group	N	Mean	Median	SD	Det.	Semi	Terr.	Flat	Freehold	New Build
Full Sample	12951289	303395	233000	344326	25.1	28.2	27.8	18.9	76.2	10.8
Flood Zone, Pre-2009	263318	263206	205000	220936	23.2	23.4	30.8	22.6	73.8	0.0
Flood Zone, Post-2009	24481	247745	201995	173422	15.4	12.8	16.6	55.3	39.8	100.0
No Flood Risk, Pre-2009	11287701	301101	228000	343633	24.1	29.2	29.3	17.4	77.5	0.0
No Flood Risk, Post-2009	1375789	330902	274995	369093	33.6	21.3	15.2	29.8	66.3	100.0

Notes: Sample restricted to English postcodes (2009–2025). "Pre-2009" and "Post-2009" groups defined by Land Registry New Build flag: Post-2009 = first sale as new build on/after 1 Jan 2009. New Build column is 100%/0% by construction.

recent construction patterns. The balance table in Table 2 examines these differences within flood zones specifically and confirms that while compositional differences exist, they are stable over time—a prerequisite for the DDD design.

Table 2: Pre-2016 Balance: Pre-2009 vs. Post-2009 Properties in Flood Zones

Variable	Pre2009	Post2009	Difference
Price	220299.29	219598.75	700.53
Log Price	12.10	12.16	-0.06
Pct Detached	23.08	12.76	10.32
Pct Semi	22.93	10.58	12.35
Pct Terraced	31.36	18.95	12.41
Pct Flat	22.64	57.71	-35.08
Pct Freehold	73.99	36.38	37.61

6. Empirical Strategy

6.1 Triple-Difference Design

I estimate the effect of Flood Re on property prices using a triple-difference specification:

$$\log(P_{ipzt}) = \beta_1(\text{Flood}_p \times \text{Post}_t \times \text{Elig}_i) + \boldsymbol{\beta}_2(\text{two-way interactions}) + \gamma_d + \delta_t + X_i'\theta + \varepsilon_{ipzt} \quad (2)$$

where P_{ipzt} is the transaction price of property i in postcode p, district z, at time t; Flood_p indicates the postcode is in a High or Medium flood risk area; Post_t indicates the transaction occurred after April 2016; Elig_i indicates the property was not sold as a new build on or after 1 January 2009 (proxy for Flood Re eligibility); γ_d and δ_t are postcode district and year-quarter fixed effects; and X_i includes property type and tenure controls.

The coefficient of interest, β_1, captures the differential change in prices for eligible

properties in flood zones after Flood Re's launch, relative to: (a) ineligible properties in the same flood zones, (b) eligible properties in non-flood areas, and (c) the baseline differential in non-flood areas. Standard errors are two-way clustered by postcode district (approximately 120 districts) and year-quarter (68 quarters in the main 2009–2025 sample). Two-way clustering accounts for both spatial correlation within districts and temporal correlation within quarters.

6.2 Identifying Assumptions

The DDD requires that, absent Flood Re, the price gap between pre-2009 and post-2009 properties would have evolved identically in flood zones and non-flood zones. This is a weaker assumption than the standard parallel trends requirement because it allows for: (1) differential trends across flood zones (absorbed by Flood × Post), (2) differential trends across construction vintages (absorbed by Post × Elig), and (3) permanent differences between eligible and ineligible flood-zone properties (absorbed by Flood × Elig).

The key threat to identification would be a shock that differentially affected pre-2009 versus post-2009 properties in flood zones but not elsewhere, coinciding with April 2016. I investigate this threat through event-study diagnostics showing no differential pre-trends, and through placebo tests using Very Low risk areas where Flood Re is irrelevant.

6.3 Event Study Specification

To visualize the timing of effects and validate the pre-trends assumption, I estimate year-specific DDD coefficients:

$$\log(P_{ipzt}) = \sum_{k \neq 2015} \beta_k (\text{Flood}_p \times \mathbb{I}[t = k] \times \text{Elig}_i) + \boldsymbol{\gamma}(\text{controls} + \text{FE}) + \varepsilon_{ipzt} \tag{3}$$

where 2015 (the year before Flood Re) is the reference year. The $\{\beta_k\}_{k=2009}^{2015}$ coefficients test the pre-trends assumption, while $\{\beta_k\}_{k=2016}^{2025}$ trace the dynamic treatment effect.

6.4 Threats to Validity

Several potential concerns deserve discussion.

Compositional changes. If Flood Re altered the types of properties being built or sold in flood zones, the DDD could reflect composition shifts rather than price effects. This concern is mitigated by including property type and tenure controls, which absorb the most important compositional differences. However, unobserved quality differences between pre-2009 and post-2009 properties—such as energy efficiency, flood resilience features in newer construction,

or interior condition—cannot be controlled for in the data. The DDD design handles time-invariant quality differences through the Flood × Eligible interaction, but any differential evolution of quality between vintages in flood zones versus non-flood zones would bias the estimate.

Anticipation. Flood Re was legislated by the Water Act 2014, two years before its operational launch in April 2016. If market participants anticipated the policy and adjusted prices accordingly, effects might appear before the operational date. The event study specification directly tests for anticipation effects in the 2014–2015 window. However, gradual anticipation beginning in 2014 would compress the pre-treatment period and potentially bias the pre-trend test.

Spillover effects. If Flood Re affected property prices in non-flood zones—for example, through general equilibrium effects in local housing markets where flood-zone and non-flood-zone properties are substitutes—the DDD control group is contaminated. The direction of bias is ambiguous: if eligible flood-zone properties became more attractive, some buyers might substitute away from non-flood properties (raising flood-zone prices and lowering non-flood prices, biasing the DDD upward) or general confidence effects could raise prices in both areas.

Vintage depreciation trends. Perhaps the most important concern is that pre-2009 and post-2009 properties may experience different price trajectories for reasons unrelated to insurance—including differential physical depreciation, energy efficiency standards (post-2006 buildings must meet stricter thermal performance requirements), flood resilience features incorporated in newer construction, and evolving buyer preferences for modern housing. If this vintage depreciation accelerates differentially in flood-classified areas (where environmental exposure may hasten deterioration of older housing stock), the DDD would capture this trend rather than the insurance subsidy. The placebo tests in Very Low risk areas are designed to detect this concern.

7. Results

7.1 Main Results

Table 3 reports the DDD estimates. The results reveal something striking: the apparent effect of Flood Re depends entirely on whether we account for what kinds of homes are being compared.

Without property controls (Column 1), eligible flood-zone homes appear to appreciate by 9.4% relative to ineligible ones—exactly what the insurance capitalization story predicts. But this vanishes the moment we control for property type and tenure (Column 2), flipping to −8.5%. This reversal reflects important compositional differences: post-2009 new builds in

Table 3: Effect of Flood Re on Property Prices: Triple-Difference Estimates

	(1)	(2)	(3)	(4)
Flood Zone × Post-2016 × Pre-2009	0.0944***	-0.0851***	-0.0143***	-0.0184***
	(0.0248)	(0.0218)	(0.0030)	(0.0040)
Two-way interactions	Yes	Yes	Absorbed	Absorbed
Property controls	No	Yes	Yes	Yes
District FE	Yes	Yes	Yes	No
Postcode FE	No	No	No	Yes
Year-Quarter FE	Yes	Yes	No	Yes
District × Year-Quarter FE	No	No	Yes	No
Observations	12,951,268	12,951,268	12,951,138	12,853,145

Notes: Standard errors clustered by postcode district and year-quarter in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Dependent variable is log transaction price. Sample restricted to English postcodes. Observation counts vary across specifications due to singleton fixed effects dropped by `fixest`.

flood zones are disproportionately flats (55% vs 23% for pre-2009) and leasehold (60% vs 26%), while pre-2009 properties are more likely to be detached or semi-detached houses. Although post-2009 properties have higher raw mean prices (reflecting the "new build premium"), controlling for property type and tenure reveals that within comparable property categories, the DDD is negative.

The most credible specifications include richer fixed effects. Column 3 adds district-by-year-quarter interactions, absorbing all local time trends, and yields -0.014 (SE: 0.003). Column 4 replaces district fixed effects with postcode-level fixed effects—the most saturated specification—and estimates -0.018 (SE: 0.004). These estimates are small in magnitude but statistically significant, implying a 1.4–1.7% relative price decline for eligible properties in flood zones after Flood Re—the opposite of theoretical predictions.

7.2 Event Study

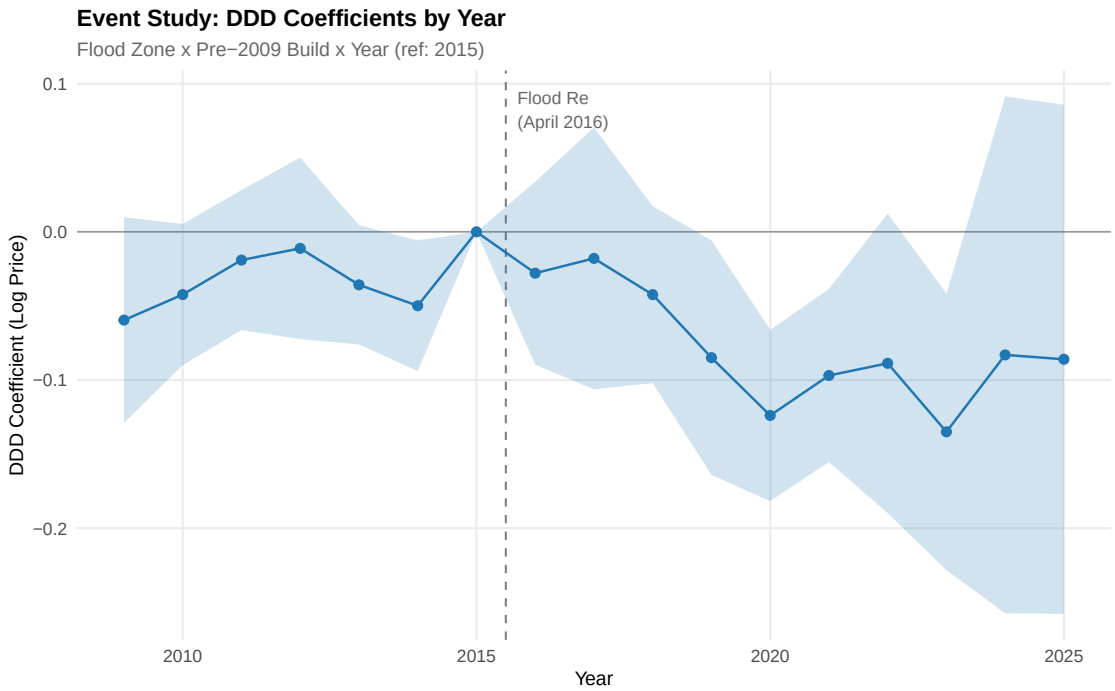

Figure 1: Event Study: DDD Coefficients by Year

Notes: Each point shows the DDD coefficient estimated from a two-period model comparing year k against the 2015 reference year, with district and year-quarter fixed effects plus property controls. Shaded band shows 95% confidence intervals. Standard errors clustered by postcode district and year-quarter. Vertical dashed line marks Flood Re launch (April 2016).

Figure 1 presents the event study, where each coefficient represents the DDD estimated using a two-year window against the 2015 reference year. Two features stand out. First, the pre-treatment coefficients (2009–2014) are all negative, with point estimates ranging from -0.060 (2009) to -0.014 (2012), and a joint test rejects the null of no pre-trends ($\chi^2(6) = 16.63$, $p = 0.011$). The confidence intervals for some years overlap zero, but the consistent negative sign suggests a secular trend. This is a meaningful identification concern: the vintage price gap was evolving differentially in flood zones even before Flood Re. Second, the post-treatment coefficients (2016–2025) are also negative and generally larger in magnitude, reaching -0.13 in 2020 and 2023. The widening gap post-2016 is consistent with a treatment effect *or* with an accelerating pre-existing trend. I apply the Rambachan and Roth (2023) sensitivity framework to formally assess the implications of the pre-trend for causal inference (see Section 7.5).

7.3 Heterogeneity

7.3.1 By Flood Risk Level

If the DDD captures insurance capitalization, the effect should be larger in higher-risk areas where the premium reduction is greater. I estimate separate DDD coefficients for High and Medium risk areas. The High-risk DDD is -0.034 (SE: 0.047) and the Medium-risk DDD is $+0.019$ (SE: 0.026)—neither is statistically significant when estimated separately. While the High-risk point estimate is more negative, the imprecision prevents meaningful dose-response inference. The absence of a clear dose-response pattern is inconsistent with straightforward insurance capitalization.

7.3.2 By Property Value

Table 4: Heterogeneous Effects by Property Type and Price Quartile

Dimension	Category	Estimate	SE	N
Property Type				
Property Type	D	-0.0065	0.0238	3250322
Property Type	S	-0.0233	0.0239	3649398
Property Type	T	-0.0914	0.0229	3599029
Property Type	F	0.0426	0.0372	2452484
Price Quartile				
Price Quartile	Q1	0.0210	0.0345	3238018
Price Quartile	Q2	0.0199	0.0057	3245624
Price Quartile	Q3	0.0188	0.0073	3270173
Price Quartile	Q4	-0.0596	0.0285	3197416

Table 4 reports DDD estimates by property price quartile and property type. The pattern is informative. By price quartile, the effect is negative only for the highest quartile (-0.057, SE: 0.028), while lower quartiles show small positive but mostly insignificant coefficients. This is the opposite of the prediction that higher Council Tax bands should benefit more from Flood Re's premium caps.

By property type, the effect is driven entirely by terraced houses (-0.092, SE: 0.023), while detached (-0.005, NS), semi-detached (-0.022, NS), and flats ($+0.044$, NS) show no significant effect. The concentration of the negative DDD in terraced houses may reflect that this property type has the largest composition shift between pre-2009 and post-2009 stock in flood zones, or that vintage depreciation is particularly acute for older terraced housing in flood-prone areas.

7.4 Placebo Tests

The placebo tests are the most informative diagnostic for the identification strategy. In Very Low flood risk areas, where standard insurance is affordable and Flood Re is irrelevant, the placebo DDD should be zero if the main estimate captures the insurance channel. Instead, the Very Low risk placebo is -0.134 (SE: 0.034, $p < 0.001$)—*larger* in magnitude than the main estimate and highly significant. The Low risk placebo is -0.033 (SE: 0.019, $p = 0.089$), marginally significant.

The failing Very Low risk placebo is a serious identification concern. It implies that the DDD is capturing a differential trend in pre-2009 versus post-2009 property prices that occurs in areas with *any* flood risk classification—not just High/Medium risk areas where Flood Re operates. This pattern is consistent with accelerating vintage depreciation in flood-classified areas (which tend to be in riparian, coastal, or lowland locations with distinct housing stock characteristics) rather than the Flood Re insurance subsidy specifically.

I therefore interpret the main DDD estimates as an upper bound on the Flood Re effect. As a rough placebo-adjusted bound, subtracting the Very Low risk placebo (-0.134) from the main DDD (-0.014) yields a residual of $+0.12$—suggesting that after netting out the confounding vintage trend, there may be a small positive insurance effect, though this crude adjustment carries substantial uncertainty. The most defensible interpretation is that the data provide evidence against large positive capitalization (the predicted 3–15% range is firmly excluded) but cannot isolate a precise Flood Re effect from confounding vintage dynamics.

7.5 Robustness

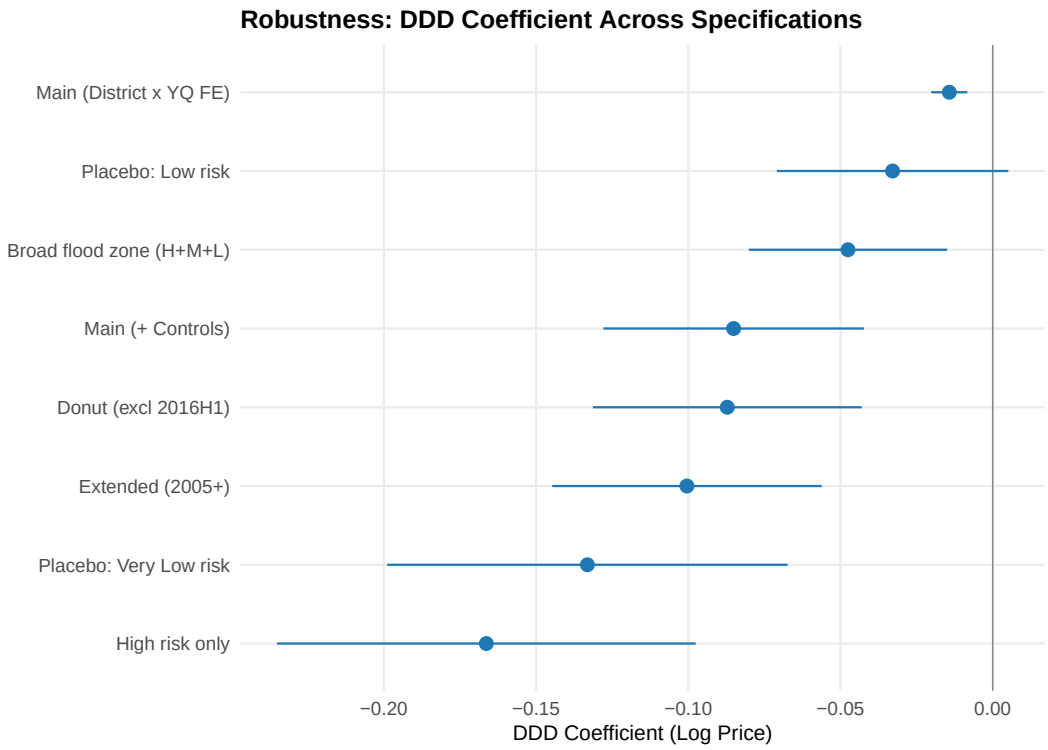

Figure 2: Robustness: DDD Coefficient Across Specifications

Notes: Each point shows the DDD coefficient from a different specification. Error bars show 95% confidence intervals. See Table 5 for details.

Figure 2 summarizes the robustness checks (full details in Table 5 in the Appendix). The negative DDD is remarkably stable: it persists across alternative flood zone definitions (High only: -0.165, SE: 0.035; High + Medium + Low: -0.048, SE: 0.017), donut estimation excluding the Flood Re transition period (-0.087, SE: 0.023), the extended pre-period sample from 2005 (-0.100, SE: 0.023), and leave-one-out analysis excluding each of the 10 largest postcode areas (Figure 4). The leave-one-out coefficients range from -0.081 to -0.096, confirming that no single region drives the result.

The Rambachan and Roth (2023) HonestDiD framework provides formal sensitivity analysis. Under exact parallel trends ($\bar{M} = 0$), the 95% robust confidence interval is [-0.089, 0.033], which includes zero. Allowing the post-treatment trend to deviate from parallel by up to the magnitude of the largest pre-trend violation ($\bar{M} = 1$) yields [-0.146, 0.090]. The HonestDiD analysis thus confirms that we cannot reject a zero effect of Flood Re on property values once pre-trend violations are accounted for.

7.6 Transaction Volume and Market Liquidity

Beyond price effects, Flood Re may have improved market liquidity by restoring insurability. I estimate the effect of Flood Re on transaction volumes at the postcode-quarter level. The DD coefficient on flood zone × post-2016 is −0.005 (SE: 0.001), indicating a small *decline* in flood-zone transaction volumes after 2016 relative to non-flood zones. This is inconsistent with the hypothesis that Flood Re restored market liquidity, though the magnitude is small and may reflect other factors affecting flood-zone markets over this period.

7.7 Moral Hazard: New Construction in Flood Zones

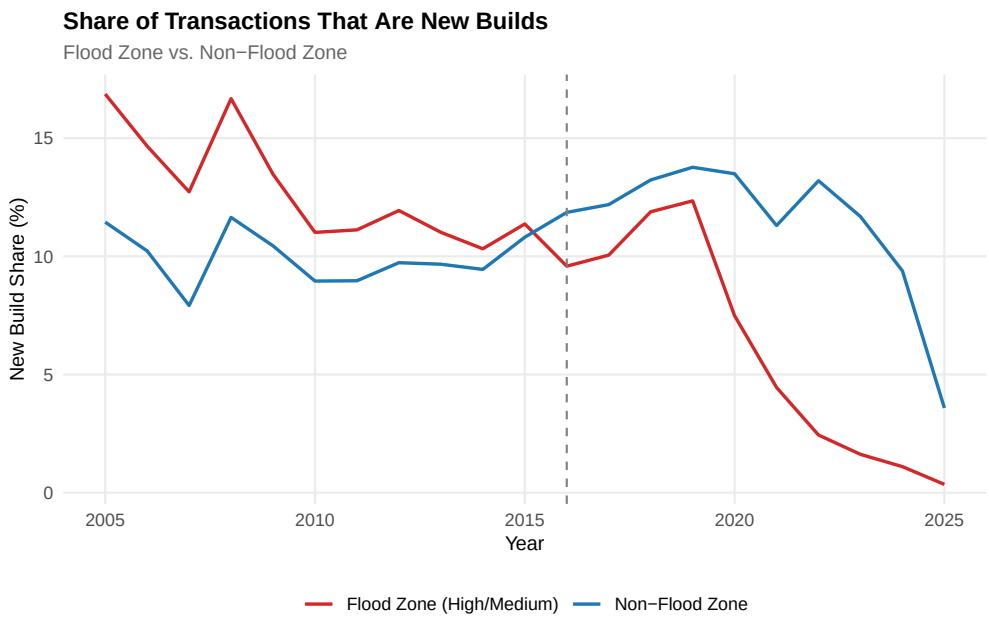

Figure 3: Share of New Build Transactions in Flood Zones vs. Non-Flood Zones

Notes: Shows the share of annual transactions that are new builds, separately for flood zones (High/Medium risk) and non-flood zones. Vertical line marks Flood Re launch (April 2016). New builds in flood zones are ineligible for Flood Re.

A central policy question is whether Flood Re's pre-2009 cutoff successfully discouraged new construction in flood zones. Figure 3 plots the new-build share over time. The DD coefficient on flood zone × post-2016 for the new-build indicator is −0.049 (SE: 0.008), indicating that the share of new construction in flood zones *fell* by 4.9 percentage points relative to non-flood zones after 2016. This is consistent with the Flood Re exclusion having a deterrent effect on flood-zone development, though it also coincides with tightened planning

regulations in flood-risk areas and the Environment Agency's strengthened objection protocol for development in Flood Zones 2 and 3.

The decline in new construction provides suggestive evidence that the market partially internalized the Flood Re exclusion—builders and buyers appear to have responded to the insurance disadvantage facing post-2009 flood-zone properties. This stands in contrast to the moral hazard predictions that dominated policy discussions before Flood Re's launch.

8. Discussion

8.1 Interpreting the Results

The absence of detectable positive capitalization admits multiple explanations, and the failing diagnostics mean some interpretations cannot be distinguished from identification failure. Several channels could genuinely attenuate capitalization even if the DDD were clean. First, *buyer inattention*: property transactions are complex, and flood insurance eligibility may not be salient to buyers who focus on price, location, and condition. Flood Re eligibility is not recorded in Land Registry data or standard property listings, so buyers must independently determine whether a property was built before 2009. If a substantial fraction of buyers are unaware of or confused about the eligibility distinction, the subsidy cannot fully capitalize. Second, *policy uncertainty*: Flood Re was designed as a transitional scheme expiring in 2039, creating uncertainty about the subsidy's duration. Rational buyers would discount the present value of an uncertain and time-limited subsidy. Third, *offsetting information effects*: Flood Re's launch may have increased awareness of flood risk in affected areas, creating a countervailing negative price effect that offset any insurance subsidy capitalization. Fourth, *supply response*: if Flood Re made existing flood-zone properties more attractive, this could have stimulated market entry by previously illiquid properties, increasing supply and muting price effects.

However, the failing placebo tests and pre-trends raise a more fundamental possibility: the DDD may simply lack the power to isolate the insurance channel from confounding vintage dynamics. The Very Low risk placebo (-0.134) is an order of magnitude larger than the main DDD (-0.014), suggesting that vintage depreciation trends in flood-classified areas dominate whatever signal the insurance subsidy generates. Under this interpretation, the "null result" reflects identification limitations rather than genuine non-capitalization.

8.2 Comparison with Prior Evidence

Garbarino et al. (2024) estimated a price increase of approximately 1.5% for flood-zone

properties using a difference-in-differences design comparing flood-zone and non-flood-zone property prices before and after Flood Re's launch. My DDD—which adds a third difference for construction vintage—finds no positive effect. This discrepancy is informative and warrants careful interpretation.

The DiD estimates the average treatment effect on all flood-zone properties relative to non-flood-zone properties. The DDD isolates the within-flood-zone insurance channel by comparing eligible and ineligible properties. The divergence between the two estimates implies that the DiD is capturing something other than (or in addition to) the insurance subsidy. Three candidate explanations exist. First, Flood Re's launch coincided temporally with the aftermath of the December 2015 floods, which may have triggered government investment in flood defenses, media attention to flood-risk communities, or shifts in buyer risk perceptions—all of which could differentially affect flood-zone property values without operating through the insurance channel. Second, the Flood Re announcement itself may have signaled that flood-risk areas would receive ongoing government support, boosting confidence in these markets independently of the premium subsidy. Third, the housing market recovery following the 2016 Brexit referendum may have differentially affected flood-zone properties if these markets had been disproportionately depressed.

The comparison also speaks to methodological choices. The DDD's additional dimension of variation provides a stronger test of the insurance mechanism specifically, but at the cost of relying on the construction vintage margin—which, as the placebo tests reveal, exhibits its own dynamics in flood-classified areas. Neither design is perfect: the DiD may overstate the insurance effect by conflating it with correlated shocks, while the DDD may understate it by attributing some genuine insurance effects to the broader vintage trend. The truth likely lies between the DiD's positive estimate and the DDD's null, suggesting that any Flood Re capitalization, if it exists, is modest.

8.3 Implications for Climate Adaptation Policy

If the evidence against large positive capitalization reflects genuine non-capitalization rather than identification limitations, the policy implications are more optimistic than the capitalization scenario originally hypothesized. These implications should be read as conditional on the causal interpretation, which the diagnostics do not fully support.

For flood-zone homeowners: If insurance subsidies do not substantially capitalize into property values, then the scheduled transition to risk-reflective pricing by 2039 would not impose concentrated capital losses on these homeowners. In the US context, the Biggert-Waters Act's attempt to eliminate NFIP subsidies was repealed partly because of projected property value declines in affected areas (Bin and Landry, 2008). The UK evidence is at

least consistent with the possibility that this form of political opposition may be weaker than policymakers feared.

For insurance markets: If Flood Re has not inflated property values, then the scheme is not generating windfall gains for sellers at the expense of buyers. The subsidy would benefit current occupants through reduced insurance costs—a flow benefit rather than a stock benefit. This distinction matters for assessing the scheme's distributional incidence.

For climate adaptation planning: The decline in new construction in flood zones suggests the eligibility cutoff is partially achieving its objective of discouraging development in at-risk areas, though the timing coincides with tightened planning regulations. If the construction response is genuine, Flood Re's design—subsidizing existing stock while excluding new construction—may offer a template for transitional disaster insurance programs that provides affordability relief without the perverse development incentives that have plagued the NFIP (Kousky and Michel-Kerjan, 2017; Browne and Hoyt, 2000).

For international policy design: As other countries consider similar schemes (Australia's Northern Australia Reinsurance Pool, launched in 2022, uses a comparable model), the UK experience offers a cautionary lesson alongside any encouragement. The evidence cannot rule out that the null result reflects low buyer awareness, measurement error in the eligibility proxy, or confounding vintage trends rather than genuine non-capitalization. Stronger micro-level data on premiums and take-up would be needed to draw firm policy conclusions.

8.4 Limitations

Several limitations qualify these findings, some of which substantially weaken the causal interpretation. *First*, the failing Very Low risk placebo (-0.134) and significant pre-trends ($p = 0.011$) indicate that the DDD is contaminated by vintage-specific trends in flood-classified areas. The estimated effect is better interpreted as evidence against large positive capitalization rather than a clean causal estimate. *Second*, the construction vintage classification using the Land Registry New Build flag identifies market segments (new-build first sales vs. everything else) rather than true Flood Re eligibility. The resulting measurement error is non-classical and systematically correlated with the new-build market—meaning attenuation toward zero is not guaranteed. Linking to EPC registers, Valuation Office records, or local authority building control data would allow a cleaner eligibility classification based on true construction year. *Third*, the flood risk classification reflects a single snapshot rather than the classification in effect at each transaction date; if risk maps were updated after major floods or defense investments coinciding with Flood Re, the treatment assignment may be endogenous. *Fourth*, I cannot observe insurance premiums or Flood Re take-up at the property level and therefore cannot estimate the first-stage effect of Flood Re on insurance costs—the

reduced-form null could reflect either non-capitalization of a real premium reduction, low take-up among transacting properties, or no premium reduction to capitalize. Aggregate first-stage evidence from Flood Re reports is suggestive but does not establish that the marginal transacted property experienced premium changes. *Fifth*, the event study uses two-year windows estimated independently rather than a jointly estimated specification with a consistent sample, which may obscure dynamics and complicate inference. *Sixth*, a repeat-sales design exploiting within-property price changes would eliminate time-invariant composition and quality confounds that the cross-sectional DDD cannot fully address.

9. Conclusion

This paper provides the first triple-difference estimate of how subsidized flood insurance affects property values, exploiting the UK Flood Re scheme's sharp exclusion of properties built after 2009. Using 13 million property transactions linked to postcode-level flood risk classifications, I compare the price evolution of eligible (pre-2009) versus ineligible (post-2009) properties in flood zones versus non-flood zones, before and after the scheme's April 2016 launch.

The most saturated specifications yield small negative DDD estimates of -1.4% to -1.8%, providing no evidence that Flood Re eligibility increased property values. However, placebo tests in low-risk areas—where the subsidy is irrelevant—produce effects of similar or larger magnitude, and pre-existing trends reject the parallel trends assumption ($p = 0.011$). The honest assessment is that the DDD cannot cleanly isolate the Flood Re insurance channel from confounding vintage depreciation trends in flood-classified areas.

Three conclusions follow, with appropriate caveats about identification. First, the evidence is inconsistent with large positive capitalization of the predicted 3–15% magnitude. A crude placebo-adjusted bound suggests any positive Flood Re effect is at most a few percentage points, well below the theoretical benchmark. If this reflects genuine non-capitalization, the political economy barriers to risk-reflective pricing transitions may be weaker than feared—in contrast to the US experience, where projected NFIP-related property value declines contributed to the partial repeal of the Biggert-Waters Act.

Second, the decline in new flood-zone construction after 2016 suggests that Flood Re's eligibility cutoff may function as a development deterrent, though the timing coincides with tightened planning regulations. If the construction response is genuine, the development decision margin responds to insurance eligibility rules even when the property value margin does not—a finding with implications for the design of future disaster insurance programs.

Third, the methodological contribution lies in demonstrating that within-flood-zone

variation in insurance eligibility can reveal confounding shocks that contaminate simpler DiD designs. The divergence between my DDD and Garbarino et al. (2024)'s positive DiD estimate illustrates the importance of controlling for concurrent flood-zone trends when evaluating insurance policies that coincide with flood events, media attention, and government investment.

The paper's limitations are substantial and point to clear directions for future work. The failing placebo tests and pre-trends mean the DDD is not a clean causal estimate. The construction vintage proxy based on the Land Registry New Build flag conflates insurance eligibility with the new-build market segment. Direct data on Flood Re take-up and premium reductions at the property level would enable a micro first-stage analysis that is currently impossible. Obtaining true construction year—from EPC registers, Valuation Office records, or local authority building control data—would allow a cleaner eligibility classification. And a repeat-sales design using properties that transact multiple times could eliminate time-invariant composition and quality confounds.

As climate change intensifies flooding globally, the design of insurance institutions becomes increasingly consequential. The UK's Flood Re scheme creates potentially powerful identifying variation for studying insurance capitalization, but exploiting it credibly requires richer data than the Land Registry alone provides. The evidence assembled here rules out the most dramatic capitalization scenarios and demonstrates that the policy question is answerable—even if answering it precisely awaits better measurement.

Acknowledgements

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Project Repository: https://github.com/SocialCatalystLab/ape-papers

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A. Data Appendix

A.1 Land Registry Price Paid Data

The Price Paid Data is published by HM Land Registry under the Open Government Licence v3.0. I download annual CSV files from the Land Registry bulk data service for years 2005–2025. Each record contains 16 fields: transaction ID, price, date, postcode, property type (D/S/T/F/O), old/new (Y/N), duration (F/L), address components, district, county, PPD category (A/B), and record status.

Sample restrictions:

1. Restrict to PPD Category A (standard residential transactions): removes additional price paid entries
2. Restrict to property types D, S, T, F: removes "Other" category
3. Require valid postcode and positive price
4. Main sample: 2009–2025 (post-2009 vintage must exist)

Construction vintage classification: A property is classified as "post-2009" (Flood Re ineligible) if Old/New = "Y" (newly built) and the transaction date is on or after 1 January 2009. All other properties—established stock (Old/New = "N") or new builds sold before 2009—are classified as "pre-2009" (Flood Re eligible). This classification is conservative: it may undercount post-2009 builds if their first transaction is recorded without the New Build flag.

A.2 Environment Agency Flood Risk Data

The "Open Flood Risk by Postcode" dataset is derived from the Environment Agency's Risk of Flooding from Rivers and Sea (RoFRS) product, which models flood probability on a 50-meter grid accounting for existing flood defenses. The grid cells are aggregated to the postcode level using Ordnance Survey address data. Each postcode is assigned the highest risk band of any grid cell within its boundary.

Risk bands:

- High: >3.3% annual probability (1 in 30 years)
- Medium: 1–3.3% annual probability (1 in 30 to 1 in 100)
- Low: 0.1–1% annual probability (1 in 100 to 1 in 1,000)
- Very Low: <0.1% annual probability (<1 in 1,000)

B. Identification Appendix

B.1 Pre-Trends

The event study (Figure 1) shows that DDD coefficients for 2009–2014 are uniformly negative, with a joint test rejecting the null of no pre-trends ($\chi^2(6) = 16.63$, $p = 0.011$). The pre-existing negative trend complicates causal interpretation; see the HonestDiD analysis in Section 6.5.

B.2 Placebo Tests

I run two placebo tests using flood risk areas where Flood Re should have no effect:

- **Very Low risk:** Properties in Very Low risk postcodes (vs. No Risk). DDD = -0.134 (SE: 0.034, $p < 0.001$).
- **Low risk:** Properties in Low risk postcodes (vs. No Risk). DDD = -0.033 (SE: 0.019, $p = 0.089$).

The Very Low risk placebo is highly significant and larger in magnitude than the main estimate, indicating that the DDD captures vintage-related trends in flood-classified areas rather than the Flood Re insurance channel. See Section 7.4 for detailed discussion.

C. Robustness Appendix

C.1 Robustness Checks Table

Table 5: Robustness Checks

Test	Coefficient	SE	Stars
Main (+ Controls)	-0.0851	0.0218	***
Main (District x YQ FE)	-0.0143	0.0030	***
Placebo: Very Low risk	-0.1331	0.0336	***
Placebo: Low risk	-0.0329	0.0194	*
Broad flood zone (H+M+L)	-0.0476	0.0166	***
High risk only	-0.1663	0.0351	***
Donut (excl 2016H1)	-0.0872	0.0225	***
Extended (2005+)	-0.1005	0.0226	***

C.2 Leave-One-Out Analysis

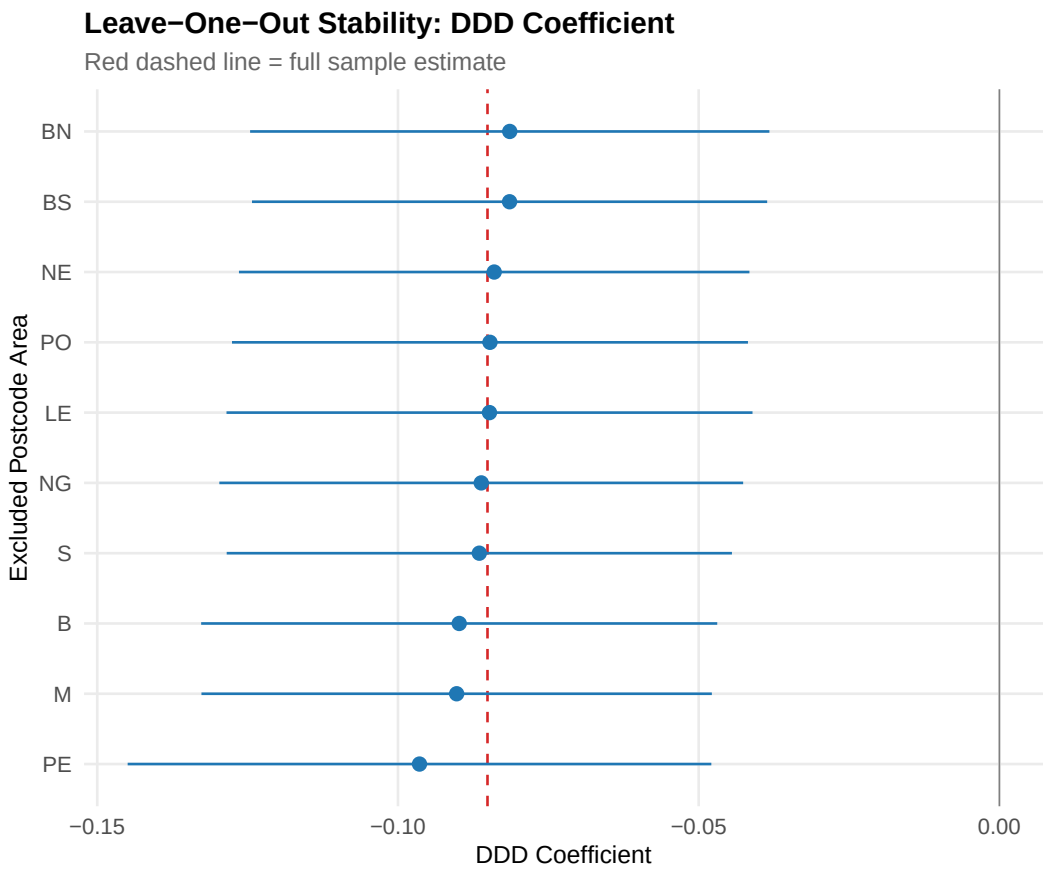

Figure 4: Leave-One-Out Stability: DDD Coefficient

Notes: Each point shows the DDD coefficient estimated after excluding one postcode area. Red dashed line shows the full-sample estimate. Error bars show 95% confidence intervals.

D. Heterogeneity Appendix

D.1 By Property Type

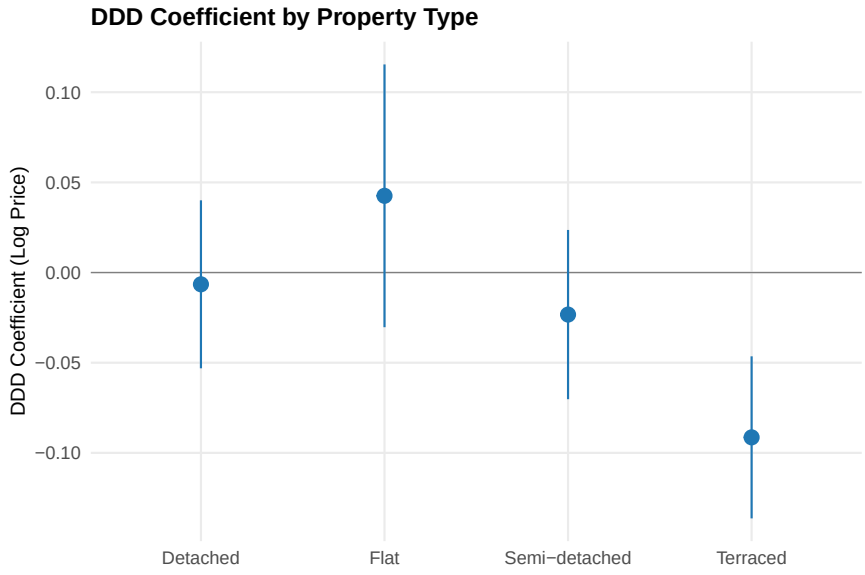

Figure 5: DDD Coefficient by Property Type

D.2 By Price Quartile

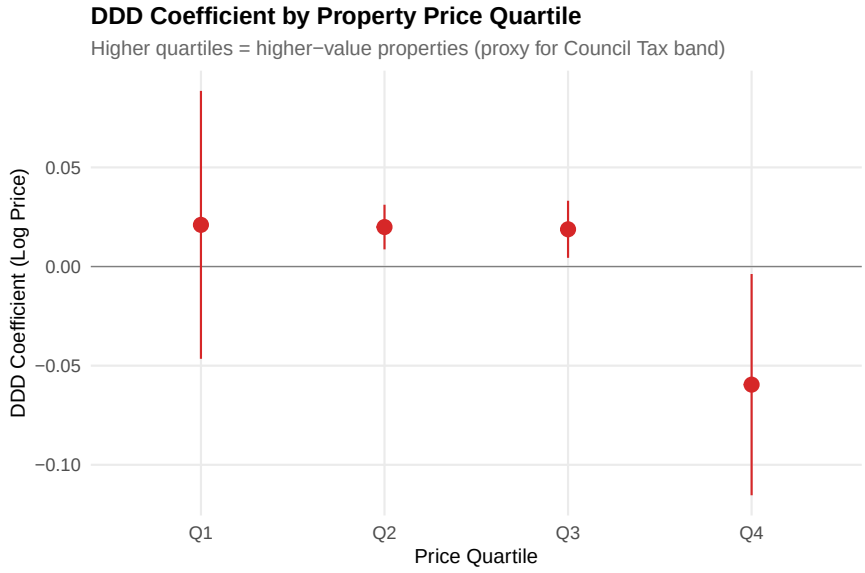

Figure 6: DDD Coefficient by Property Price Quartile

E. Additional Figures and Tables

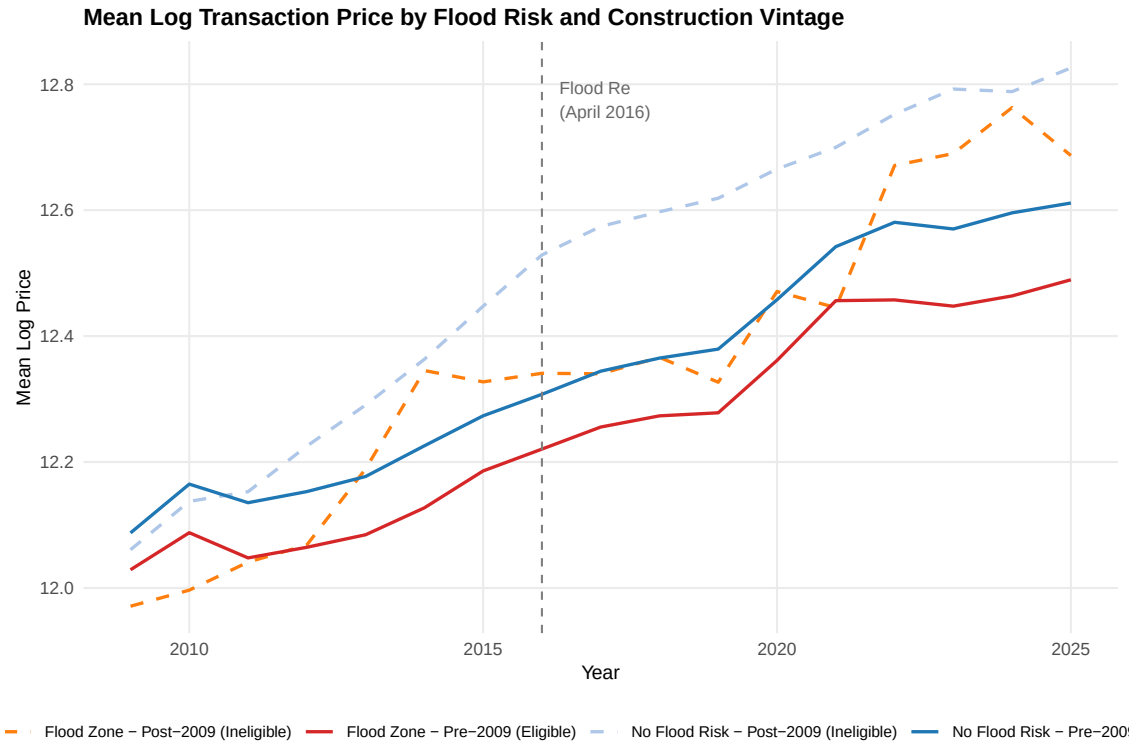

Figure 7: Mean Log Transaction Price by Flood Risk and Construction Vintage

Notes: Shows annual mean log transaction price for four groups: flood zone eligible (pre-2009), flood zone ineligible (post-2009), non-flood eligible, and non-flood ineligible. Vertical dashed line marks Flood Re launch (April 2016).

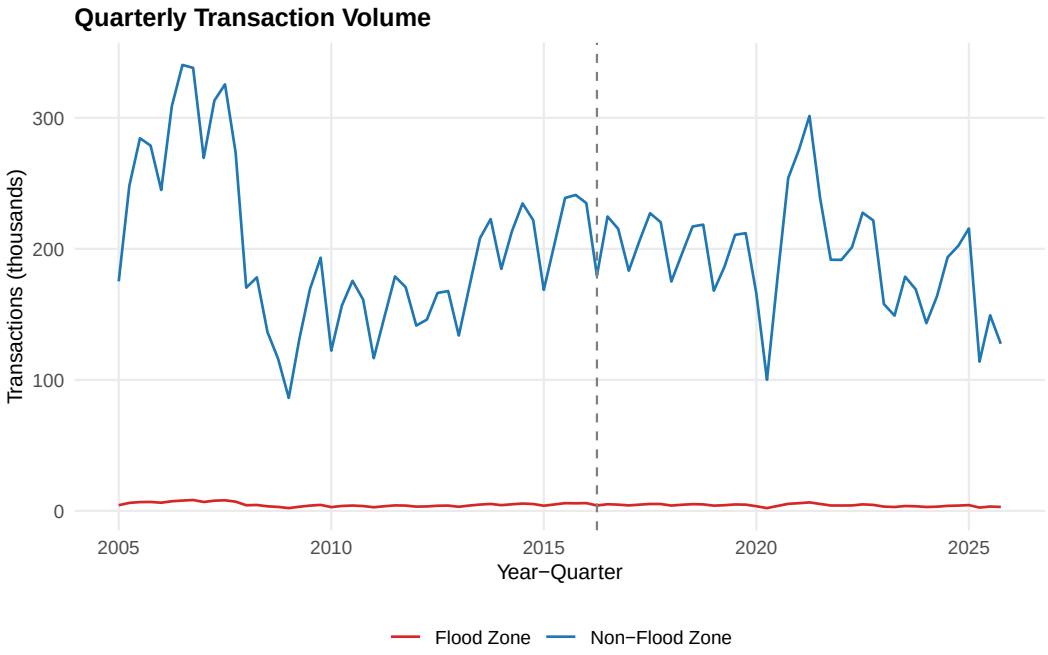

Figure 8: Quarterly Transaction Volume: Flood Zone vs. Non-Flood Zone

Notes: Shows quarterly transaction counts (thousands) for properties in flood zones (High/Medium risk) and non-flood zones. Vertical dashed line marks Flood Re launch (April 2016).